MINUTES OF MEETING (MOM)
Team 8
3D Printing Cost Estimation Platform

Meeting Details

Project:	3D Printing Cost Estimation Platform
Week:	Week 7
Date:	Friday, March 6, 2026
Time:	4:00 PM IST
Location:	CIE, Ground Floor
Client:	Anil Kumar Upadhyay, Five Fingers Innovative Solutions
Client Email:	`aku1974india@gmail.com`
Client Phone:	9949497560

Team Members Present

- Abhiroop Pullepu
- Mehrish Khan
- Ronith Menneni
- Sai Hasini
- Ruschita Guddeti

Notes

- This meeting was held one week after the Release 1 demo.
- Focus: client feedback on Release 1, small post-release fixes, and identifying priorities for the next sprint.

1 Release 1 Demo Feedback

1.1 Positive Feedback

- End-to-end flow (upload → slice → view → estimate) worked smoothly during the demo.
- Client appreciated the clear cost breakdown and the structured release notes.
- Documentation quality (README, API docs, diagrams) was called out as a strong point.

1.2 Areas for Improvement

- UI could provide more prominent warnings for models that are close to printer volume limits.
- Need clearer messaging when slicing fails due to geometric issues (thin walls, non-watertight meshes).
- Desire for additional security hardening and better admin controls in upcoming releases.

2 Post-Release Bug Fixes & Tweaks

2.1 Frontend Fixes

- **Bug: Model dimension text overlap**
  - Symptom: On smaller laptop screens, model dimension text overlapped with the 3D viewer controls.
  - Fix: Moved dimensions into a responsive side panel; added CSS breakpoints for small viewports.
- **Bug: Cost breakdown flicker**
  - Symptom: Cost breakdown card briefly showed "0" values before data loaded.
  - Fix: Added skeleton/loading state so values appear only after successful API response.

2.2 Backend Fixes

- **Bug: Failure log missing slicing parameters**
  - Symptom: Some failure log entries did not capture the exact slicing parameters used.
  - Root cause: Parameters were not always passed through to the logging helper.
  - Fix: Normalised failure logging so all code paths include job ID, filename, and full parameter JSON.
- **Bug: Idle database connections**
  - Symptom: After running for some time, occasional warnings about stale connections.
  - Fix: Tuned SQLAlchemy connection pool recycle/timeout settings based on observed usage.

3 Roadmap Discussion (Post Release 1)

3.1 Short-Term Focus (Next 2–3 Weeks)

- Improve printability analysis (better thin-wall and overhang detection, fewer false positives).
- Enhance admin panel to allow editing of more configuration values (e.g., waste/failure factors).
- Investigate STEP-to-mesh conversion options for proper STEP preview.

3.2 Medium-Term Ideas

- Introduce stronger login discipline (multi-step login, audit logs).
- Add basic job history and filtering for repeat customers.
- Expand reporting (e.g., total material used per month, job counts).

4 Action Items

Team Deliverables (Due: March 13, 2026)

Owner	Task
Abhiroop Pullepu	Refine failure logging to always capture slicing parameters and error types; add one integration test for failure log API.
Mehrish Khan	Investigate tuning of cost engine for edge cases (very small or very large models) and document any assumptions.
Ronith Menneni	Implement responsive layout fixes for the 3D viewer and dimension panel on smaller screens.
Sai Hasini	Improve user-facing error messages for slicing failures and add consistent toast notifications.
Ruschita Guddeti	Draft a high-level roadmap document for "Release 1.1" incorporating client feedback.

Client Deliverables (Due: March 13, 2026)

- Provide a small set of "problematic" STL/STEP files commonly seen in production (thin walls, large overhangs, near volume limits) for testing.

Minutes prepared by: Team 8
Date: March 6, 2026